

TRANSACTION COST EQUATIONS

This section describes the trading cost equations that will be used to solve the portfolio trader's dilemma. When performing portfolio optimization it is most beneficial to express the trading strategy in terms of a trade schedule to allow us to most effectively manage total portfolio risk and express costs in total dollars to allow us to easily sum costs across stocks. The formulations used here are based on the equations provided in Chapter 7, Advanced Algorithmic Forecasting Techniques.

For a portfolio of m stocks that are to be executed over n trading periods we have:

The trade schedule x as an $m \times n$ matrix as follows:

$$x = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix}$$

The residual schedule r as an $m \times n$ matrix as follows:

$$r = \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{pmatrix}$$

where $r_{ij} = \sum_{k=j}^n x_{ik}$.

For simplicity of notation, we define x_k and r_k to be the column vectors of the trade and residual matrices, respectively, as follows:

$$x_k = \begin{pmatrix} x_{1k} \\ x_{2k} \\ \vdots \\ x_{mk} \end{pmatrix} \quad r_k = \begin{pmatrix} r_{1k} \\ r_{2k} \\ \vdots \\ r_{mk} \end{pmatrix}$$

The covariance matrix C is an $m \times m$ matrix as follows:

$$C = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1m} \\ c_{21} & c_{22} & \cdots & c_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mm} \end{pmatrix}$$